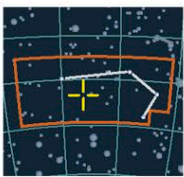


DISRUPTED SPIRAL GALAXY

Cartwheel Galaxy



CATALOG NUMBER
ESO 350-G40

DISTANCE
500 million light-years

DIAMETER
150,000 light-years

MAGNITUDE 19.3

SCULPTOR

If the Cartwheel Galaxy looks unusual, it's because it is the victim of an intergalactic "hit-and-run." The Cartwheel was once a normal spiral galaxy. As we see the galaxy, it is recovering from a head-on collision with a smaller runaway galaxy many millions of years earlier in its history. Such events are rare in the cosmos—galactic collisions usually involve grazing encounters or a slow dance toward an eventual merger. The Cartwheel shows what happens when two galaxies pass

CLOUDS IN THE CORE

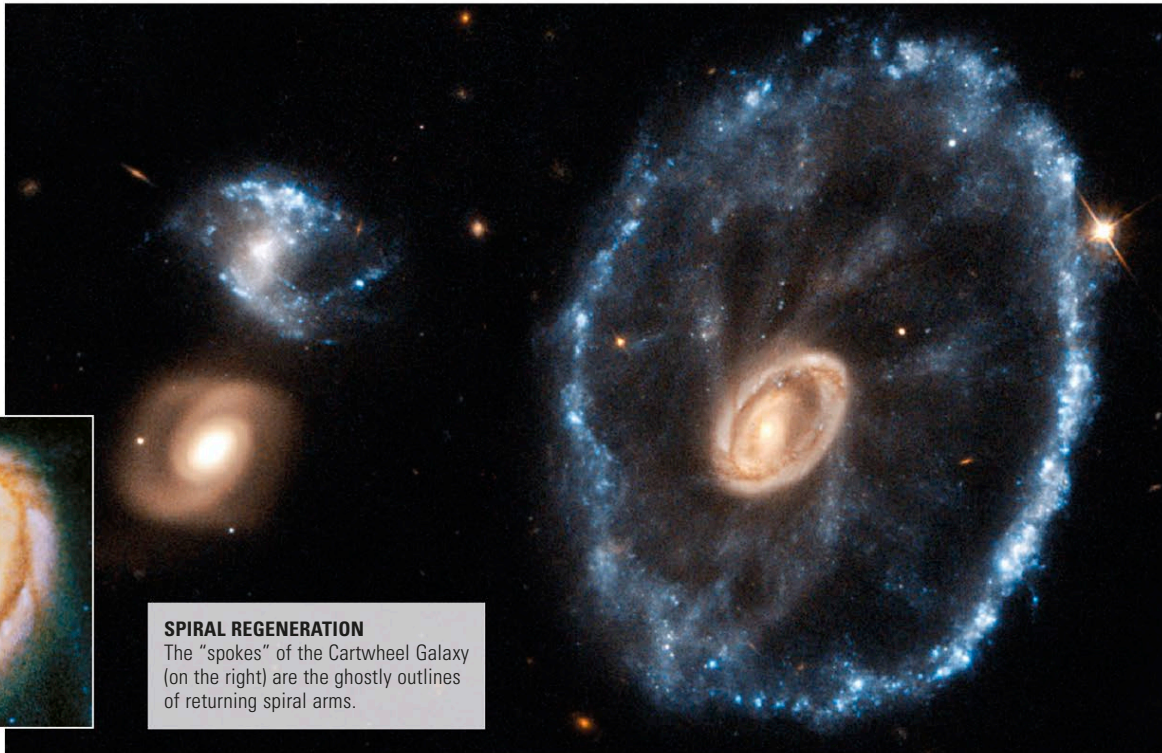
So-called "comet clouds," each a thousand light-years long, are found in the Cartwheel's core. They are thought to arise as hot, fast-moving gas set in motion by the collision plows through denser, slower-moving matter.



through each other at high speed while oriented at right angles to each other. The rotating density wave that is normally responsible for the spiral arms was disrupted in this case, resulting in the disappearance of the spiral structure. Meanwhile, a shock wave spread to the outer edge of the galaxy, creating a ring of vigorous star

formation. An inward-traveling shock wave is probably responsible for the core's unusual "bull's-eye" appearance. For years, most astronomers suspected that one of the Cartwheel's two immediate neighbors was responsible for the collision. Both showed signs of being the culprit—a nearby small, blue galaxy has a

disrupted shape and vigorous star formation, while a yellow galaxy could have been stripped of its star-forming gas in the encounter. However, recent radio observations have shown a telltale stream of gas leading from the Cartwheel toward another small galaxy a quarter of a million light-years away.

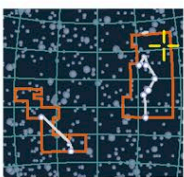


SPIRAL REGENERATION

The "spokes" of the Cartwheel Galaxy (on the right) are the ghostly outlines of returning spiral arms.

RING GALAXY

Hoag's Object



CATALOG NUMBER
PGC 54559

DISTANCE
500 million light-years

DIAMETER
120,000 light-years

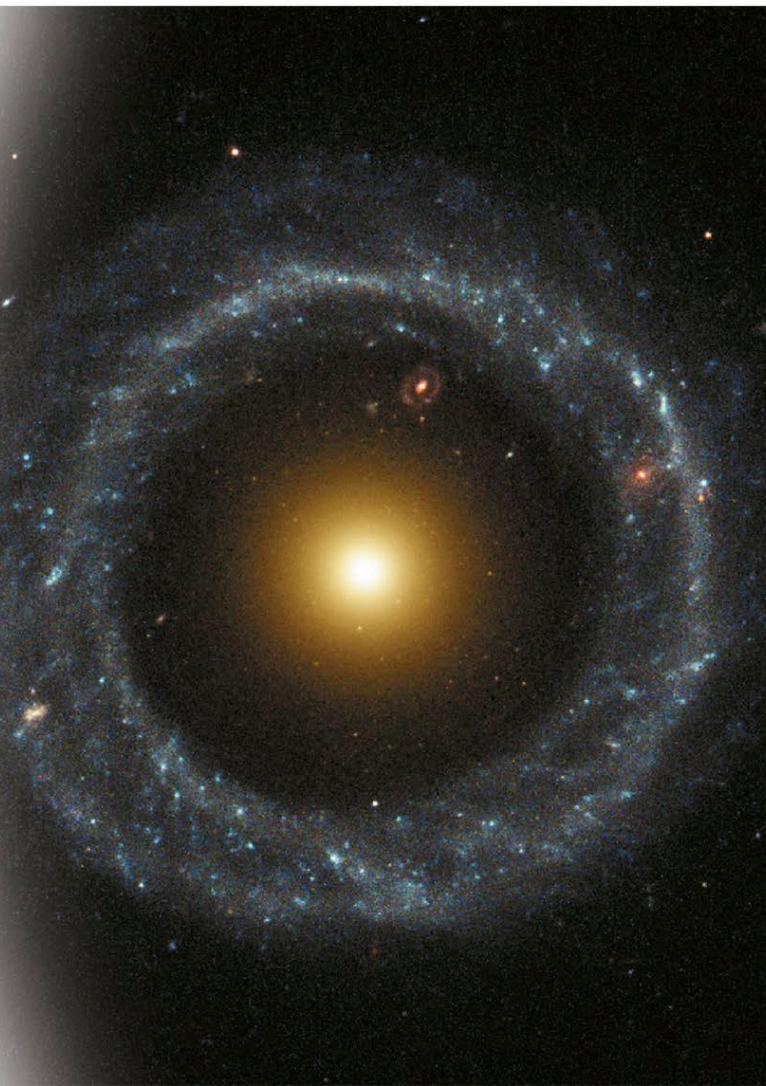
MAGNITUDE 15.0

SERPENS

Hoag's Object is one of the most bizarre galaxies in the sky. Although its ring structure suggests parallels to the Cartwheel Galaxy (a spiral disrupted by a head-on collision, see above), there are no nearby galaxies that could have caused an impact. One of two theories might account for the shape of Hoag's Object and that of similar ring galaxies. The galaxies may be members of an unusual class of spiral in which the two arms develop into a circle. Alternatively, they may be former elliptical galaxies that have each swallowed another galaxy, creating a surrounding ring of star-forming material.

SEE-THROUGH GALAXY

The gap between Hoag's Object's core and its ring is truly transparent—a background galaxy can be seen through it near the top of this image. However, the gap could still contain large numbers of faint stars.



LOW-SURFACE-BRIGHTNESS GALAXY

Malin 1



CATALOG NUMBER
None

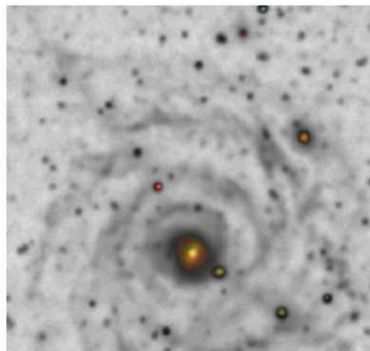
DISTANCE
1 billion light-years

DIAMETER
600,000 light-years

MAGNITUDE 25.7

COMA BERENICES

Despite its dull appearance, Malin 1 is an extremely important galaxy. Discovered by accident in 1987, it is an enormous but faint spiral that is for some reason poor at forming stars. It seems that such low-surface-brightness galaxies could account for up to half the galaxies in the universe, though Malin 1 is one of the largest of the type.



MALIN 1 IN A NEGATIVE IMAGE